ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

(i) Pixel pitch p = 5 microm = 0.005 mm. Hence:

$$f_x = f/p = 16/0.005 = 3200 pixels$$

(ii) Sensor width = 1280(0.005) = 6.40 mm. At standoff Z = 400 mm, belt width W = Z(6.40)/16 = 160 mm.

$$Scale = 160/1280 = 0.125 mm/pixel$$

Working: $5\,\mu$m $= 5 \times 10^{-3}$ mm $= 0.005$ mm. So $16/0.005 = 16 \times 200 = 3200$ pixels. For the sensor, $1280 \times 0.005 = 6.4$ mm. Then $W = (400/16) \times 6.4 = 25 \times 6.4 = 160$ mm, and $160/1280 = 0.125$ mm/pixel.

Q1(b) [3 marks]

ADC step = 5/2^12 = 0.0012207 V = 1.2207 mV. Sensor sensitivity is 10 mV/N, so ΔF = 1.2207/10 = 0.12207 N. The smallest resolvable force change is approximately 0.122 N.

Calculation: $2^{12} = 4096$ levels. One step is $5/4096 = 0.0012207$ V, i.e. 1.2207 mV. Dividing by the sensitivity, $1.2207\,\text{mV}/(10\,\text{mV/N}) = 0.12207$ N. The mV units cancel, leaving N.

Q1(c) [3 marks]

For zero-order hold, F = e^(AT) and G = integral_0^T e^(Aτ)B dτ. Here:

$$A = -2, B = 1, T = 0.1s$$

Thus:

$$F = e^(- 0.2) = 0.818731$$

$$G = (1 - e^(- 0.2))/2 = 0.0906346$$

Therefore:

$$x_{k+1} = 0.818731 x_k + 0.0906346 u_k$$

With $AT = (-2)(0.1) = -0.2$, $F = e^{-0.2} = 0.8187308$. For G, $(1 - F)/2 = (1 - 0.8187308)/2 = 0.1812692/2 = 0.0906346$. Thus the input contribution is positive.

Question 2 [10 marks]

Q2(a) [2 marks]

Choose $x_1 = \theta$ and $x_2 = \dot{\theta}$. Then $\dot{x}_1 = x_2$ and $\dot{x}_2 = [\tau - b\, x_2 - mg\ell \sin(x_1)]/J$. At equilibrium:

$$x_2 = 0 and_0 = mg\ell sin\theta_e$$

Since $|\sin \theta_e|$ less than or equal to 1, an equilibrium exists when $|\tau_0|$ less than or equal to $mg\ell = 2.4525$ N m.

At rest $\dot{\theta} = 0$ and $\ddot{\theta} = 0$, so the damping term vanishes. Rearranging gives $\tau_0 = mg\ell \sin\theta_e$. Here $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N m. Since $-1 \leq \sin\theta_e \leq 1$, the required torque must lie between -2.4525 and $+2.4525$ N m.

Q2(b) [2 marks]

The Jacobian has $A_{12} = 1$ and $A_{22} = -b/J = -2.5$. The input matrix is $B = [0, 50]^T$.

Q2(c) [4 marks]

At rest, $\dot{\theta} = \ddot{\theta} = 0$, so the torque balance is $Kp(\theta d - \theta ss) = mg\ell \sin \theta ss$. With $Kp = 50$, $\theta d = \pi/2$ and $mg\ell = 2.4525$, solve $50(\pi/2 - \theta ss) = 2.4525 \sin \theta ss$. This gives θss approximately 1.52181 rad = 87.193°, so the error is 90°-87.193° = 2.807°; it is not zero.

Rearranging the steady-state equation gives $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. I used radians in the equation. Solving numerically gives $\theta_{ss} \approx 1.522$ rad $\approx 87.19°$, so the gap from $90°$ is about $2.81°$.

Q2(d) [2 marks]

The added gravity torque should cancel the load, so the steady-state error becomes zero.

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

For $H(s)=\omega c/(s+\omega c)$, $\omega c = 2\pi(20) = 125.664$ rad/s and $T = 0.001$ s. Tustin gives $H(z) = \alpha(1+z^{-1})/[(1+\alpha)+(\alpha-1)z^{-1}]$, $\alpha = \omega cT/2 = 0.0628319$. Thus:

$$y[k] = 0.881765y[k-1] + 0.0591175(x_k + x[k-1])$$

DC gain = (2x0.0591175)/(1-0.881765) approximately 1.000.

$\omega_c = 2\pi(20) = 125.664$ rad/s. With $T = 0.001$ s, $\alpha = \omega_c T/2 = 125.664(0.001)/2 = 0.062832$. The recurrence coefficients are approximately 0.881765 on the previous output and 0.0591175 on each input sample. Check: $2(0.0591175)/(1 - 0.881765) \approx 1$, so the DC gain is close to one.

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. The desired polynomial is $(s+10)^2 = s^2+20s+100$.

Q3(d) [2 marks]

An observer uses the plant model and known control input to estimate unmeasured states such as velocity, and its poles can be designed. A simple low-pass filter cannot exploit the state-space model in this way. A filter may be safer when the model is poorly known or unreliable, because it relies less on model accuracy.

The observer uses the plant model and the measured output to reconstruct the unmeasured state. Its convergence depends on the model and the observer poles. A low-pass filter is simpler, but generally introduces phase lag and does not explicitly use the plant dynamics.

Question 4 [10 marks]

Q4(a) [2 marks]

Closed-loop characteristic polynomial is $s^2+(a+bk_2)s+(c+bk_1)=s^2+4s+8$, giving poles -2±2j. At DC, $\theta/r = bN_{ff}/(c+bk_1)$, so unity gain requires $N_{ff}=(c+bk_1)/b=(6+2)/1=8$.

Comparing coefficients with $s^2 + 4s + 8$ gives $a + bk_2 = 4$ and $c + bk_1 = 8$. The feedforward gain is chosen from the nominal plant so the reference scaling is unity; here this gives $N_{ff} = 8$. The closed-loop roots of $s^2 + 4s + 8$ are $-2 \pm 2j$.

Q4(b) [2 marks]

$$\theta_{ss} = 0.75 \times 8/(6 + 2) = 0.75,$$

the percentage tracking error is 25%

Q4(c) [4 marks]

With $\dot{\xi}$=r-θ and $u=-k_1\theta-k_2\dot{\theta}+K_I\xi$, the homogeneous closed-loop characteristic polynomial is $s^3+(a+bk_2)s^2+(c+$
Desired polynomial: $(s+4)[(s+2)^2+4]=(s+4)(s^2+4s+8)=s^3+8s^2+24s+32$. For a=2,b=1,c=6, matching coefficients gives k_2=6, k_1=18 and K_I=32.

First multiply $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$. Comparing this with the augmented characteristic polynomial gives $k_2 = 6$, $k_1 = 18$, and $K_I = 32$. I checked the coefficients in descending powers of s.

Q4(d) [2 marks]

At steady state, a stable closed loop with integral action drives $\dot{\xi}$=r-θ to zero; therefore r-θ=0 and the tracking error is zero, assuming no saturation or other implementation limits prevent convergence. A practical failure mode is integrator windup during actuator saturation; the standard remedy is anti-windup, such as back-calculation or conditional integration.

With integral action, a constant tracking error keeps accumulating in the integral state, so the controller can adjust its output until the error goes to zero, assuming the closed loop is stable

and the actuator is not constrained. Under saturation the integral state can keep growing; anti-windup limits or back-calculates that state.